

EE1200 - Basic Lab Debug Guidelines

STATUS: CIRCUIT NOT WORKING?

⚠ DO NOT PANIC.

Most errors are caused by loose wires or incorrect power settings. Follow this order.

THE GOLDEN DEBUG ORDER:

1) Power → 2) Ground → 3) Input → 4) Output → 5) Theory

1. Power Check (The #1 Culprit)

Always do this first. Do not assume the power supply is correct.

- **Is it ON?** Check the actual output button/indicator on the power supply.
- **Check Current Limit:** Is the "CC" (Constant Current) light on? If yes, you have a short circuit. Turn the current knob up slightly or find the short.
- **Measure at the Pins:** Don't measure the power supply display. Use your DMM to measure voltage **directly at the IC pins** or breadboard power rails.
- **Voltage Levels:** Is the expected voltage present? (e.g., +5V for TTL, ±12V for Op-Amps).

2. The "Common Ground" Check

- **Unified Ground:** Ensure the **Power Supply Ground**, **Function Generator Ground**, and **Oscilloscope Ground** clips are all connected to the same node (GND) on your circuit.
- **Floating Signals:** Without a common ground, your signals will look like random noise or float up/down.

3. Breadboard & Physical Inspection

- **The Rail Trap:** Check your breadboard power rails (the blue/red lines). Many breadboards have a **break in the middle**. You may need to bridge it with a wire.
- **The Tug Test:** Gently tug your wires. If a wire comes out easily, the connection was bad.
- **IC Orientation:** Is the chip facing the right way? (The notch or dot usually marks Pin 1).
- **Continuity Check:** Use the DMM "Beep" mode (Continuity) to verify connections between non-adjacent nodes. **(Turn Power OFF first!)**.

4. Signal Source (Function Generator)

- **Output Enabled:** Is the **OUTPUT** button lit up?
- **Impedance Check (Hi-Z vs 50Ω):** If your voltage is double what you expect, check the generator's load setting.
- **Cable Verification:** Connect the BNC cable directly from the Generator to the CRO (Oscilloscope) to prove the signal exists *before* connecting it to your circuit.

5. Oscilloscope (CRO/DSO) Setup

- **Probe Attenuation (1X vs 10X):** Check the switch on the physical probe handle. Ensure the Oscilloscope channel menu matches this setting (usually 10X).
- **Ground Clip:** Is the ground alligator clip actually connected to the circuit ground?
- **Coupling:** Set to **DC Coupling** mostly. (AC coupling hides DC offsets).
- **Triggering:** If the wave is scrolling wildly, set the Trigger Source to your active channel and the Mode to **AUTO**.
- **Scale:** Adjust **Volts/Div** and **Time/Div**. You might just be zoomed in too far.

6. DMM Debug Steps

- **Correct Ports:** Are your probes in the Voltage/Ohms jack or the Current (Amps) jack?
- **Mode Check:** Are you on DC Voltage or AC Voltage?
- **Resistance:** Never measure resistance while the circuit is powered on.
- **Blown Fuse:** If measuring Current (mA) reads **0.00**, the internal DMM fuse might be blown.

7. The "Divide and Conquer" Method

Do not stare at the whole circuit at once.

1. **Check Input:** Is the signal entering the first component correctly?
2. **Check Halfway:** Pick a node in the middle of the circuit. Is the signal correct there?
3. **Check Output:** Is the signal present at the final pin?
4. *If the signal is good at the Input but bad at the Halfway point, the problem is in the first half.*

Common Beginner Mistakes

- **No Common Ground:** Causes noise and floating signals.
- **Floating Inputs:** Leaving unused digital pins (like Reset or Clock) disconnected. Tie them to Vcc or Ground as per the datasheet.

- **Resistor Mix-up:** confusing $1\text{k}\Omega$ (Brown-Black-Red) with $10\text{k}\Omega$ (Brown-Black-Orange).
 - **Wrong Pinouts:** Assuming Pin 14 is always Vcc and Pin 7 is always Ground. Check the datasheet!
 - **Bypassing the Rail:** Plugging chips backward or spanning the wrong rows.
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The issue is still persisting; please contact the lab staff and TAs to resolve it.